4: Eq. differentielle.

Ex 32 :

$ay'' + by' + cy = f(x)$
$\Leftrightarrow y'' + \frac{b}{a}y' + \frac{c}{a}y = \frac{f(x)}{a}$

1° $4y'' - 5y' + 6y = e^{2t}$ (E)

Soit $(E_h) : y'' - 5y' + 6y = 0$

L'équation caractéristique:

$r^2 - 5r + 6 = 0$

$\Delta = 25 - 24 = 1 > 0$

$r_1 = \frac{5+1}{2} = 3$

$r_2 = \frac{5-1}{2} = 2$

Donc $r^2 - 5r + 6 = (r-3)(r-2)$

On en déduit : $y_h(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x}$
$= C_1 e^{3x} + C_2 e^{2x}$
$C_1, C_2 \in \mathbb{R}$

Recherche d'une solution particulière

$y_p(x) = C_1(x) e^{3x} + C_2(x) e^{2x}$
$= C_1(x) y_1(x) + C_2(x) y_2(x)$

cherchons donc $C_1(x)$ et $C_2(x)$ tq $y_p(x)$ soit solution de l'équation (E).

$$\begin{cases} C_1'(x) y_1(x) + C_2' y_2(x) = 0 \\ C_1'(x) y_1'(x) + C_2'(x) y_2'(x) = \frac{e^{2x}}{4} = e^{2x} \end{cases}$$

$$W = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = \begin{vmatrix} e^{3x} & e^{2x} \\ 3e^{3x} & 2e^{2x} \end{vmatrix}$$

$= 2e^{3x}e^{2x} - 3e^{3x}e^{2x}$
$= 2e^{5x} - 3e^{5x} = -e^{5x}$

$$C_1' = \frac{\begin{vmatrix} 0 & e^{2x} \\ e^{2x} & 2e^{2x} \end{vmatrix}}{W} = \frac{-e^{4x}}{-e^{5x}} = e^{-x}$$

$C_1(x) = \int C_1'(x)\,dx = \int e^{-x}\,dx = -e^{-x}$

$$C_2'(x) = \frac{\begin{vmatrix} e^{3x} & 0 \\ 3e^{3x} & e^{2x} \end{vmatrix}}{W} = \frac{e^{5x}}{-e^{5x}} = -1$$

$C_2(x) = \int -1\,dx = -x$

Conclusion la solution de l'équation (E) est de la forme

$y(x) = C_1 e^{3x} + C_2 e^{2x} + (-e^{-x})e^{3x} + (-x)e^{2x}$
$= C_1 e^{3x} + C_2 e^{2x} - e^{2x} - x e^{2x}$
$= C_1 e^{3x} + e^{2x}(C_2 - 1 - x)$

$$\boxed{y(x) = C_1 e^{3x} + (\alpha - x) e^{2x}}$$

$y'' - 4y' + 4y = e^{2t}$ (E)

Soit $(E_h) : y'' - 4y' + 4y = 0$

L'équation caractéristique:

$r^2 - 4r + 4 = 0$
$\Leftrightarrow (r-2)^2 = 0$

Donc la solution de (E_h)

$y_h(x) = (\alpha x + \beta) e^{2x}$

Variation de la constante.

$y_p(x) = (\alpha(x) x + \beta(x)) e^{2x}$
$= x\alpha(x) e^{2x} + \beta(x) e^{2x}$
$= \alpha(x) y_1(x) + \beta(x) y_2(x)$

avec $y_1(x) = x e^{2x}$ et $y_2(x) = e^{2x}$

$$\begin{cases} \alpha'(x) y_1(x) + \beta'(x) y_2(x) = 0 \\ \alpha'(x) y_1'(x) + \beta'(x) y_2'(x) = e^{2x} \end{cases}$$

$$\Leftrightarrow \begin{cases} \alpha'(x) x e^{2x} + \beta'(x) e^{2x} = 0 \\ \alpha'(x)(1+2x) e^{2x} + \beta'(x) 2 e^{2x} = e^{2x} \end{cases}$$

$$\Leftrightarrow \begin{cases} x\alpha'(x) + \beta'(x) = 0 \\ (1+2x)\alpha'(x) + 2\beta'(x) = 1 \end{cases}$$

$$W = \begin{vmatrix} x & 1 \\ 1+2x & 2 \end{vmatrix} = 2x - 1 - 2x = -1$$

$$\alpha'(x) = \frac{\begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix}}{-1} = \frac{-1}{-1} = 1$$

$\alpha(x) = \int \alpha'\,dx = \int 1\,dx = x$

$$\beta'(x) = \frac{\begin{vmatrix} x & 0 \\ 1+2x & 1 \end{vmatrix}}{-1} = \frac{x}{-1} = -x$$

$\beta(x) = \int -x\,dx = \frac{-x^2}{2}$

Conclusion La solution de l'équation (E) est: $y(x) = (\alpha x + \beta) e^{2x} + x^2 e^{2x} - \frac{x^2}{2} e^{2x}$

$= (\alpha x + \beta + x^2 - \frac{x^2}{2}) e^{2x}$

$= (\frac{x^2}{2} + \alpha x + \beta) e^{2x}, \alpha, \beta \in \mathbb{R}$

CamScanner

29 $y'' + y = t^2$ (E)

(E_h) : $y' + y = 0$

L'eq caract. $r^2 + 1 = 0 \Leftrightarrow r = \mp i$

$Re(r) = 0$

$Im(r) = 1$

Donc les solutions de (E_h) :

$y_h(t) = (\alpha \cos(1t) + \beta \sin(1t))\, e^{0t}$

$= (\alpha \cos(t) + \beta \sin(t))$

Recherche d'une y_p : variation de la constante

$y_p(t) = \alpha(t) \cos(t) + \beta(t) \sin(t)$

$$\begin{cases} \alpha'(t) . \cos(t) + \beta'(t) \sin(t) = 0 \\ \alpha'(t)(-\sin(t)) + \beta'(t) \cos(t) = t^2 \end{cases}$$

$$W = \begin{vmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{vmatrix} = \cos^2(t) + \sin^2(t) = 1$$

$$\alpha'(t) = \frac{\begin{vmatrix} 0 & \sin(t) \\ t^2 & \cos(t) \end{vmatrix}}{1} = -t^2 \sin(t)$$

$$\beta'(t) = \frac{\begin{vmatrix} \cos(t) & 0 \\ -\sin(t) & t^2 \end{vmatrix}}{1} = t^2 \cos(t)$$

$\alpha(t) = \int -t^2 \sin(t)\, dt$

$= (at^2 + bt + c)(\alpha \sin(t) + \beta \cos(t))$

Soit $F(t) = (at^2 + bt + c)(\alpha \sin(t) + \beta \cos(t))$

et $F'(t) = -t^2 \sin(t)$

$F'(t) = (2at + b)(\alpha \sin(t) + \beta \cos(t)) +$
$(at^2 + bt + c)(\alpha \cos(t) - \beta \sin(t))$
$= \sin(t)(\alpha 2at + \alpha b - \beta at^2 - \beta bt - \beta c)$
$+ \cos(t)(\beta 2at + \beta b + \alpha at^2 + \alpha bt + \alpha c)$
$= -t^2 \sin(t)$

$$\begin{cases} 2\alpha at + \alpha b - \beta at^2 - \beta bt - \beta c = -t^2 \\ 2\beta at + \beta b + \alpha at^2 + \alpha bt + \alpha c = 0 \end{cases}$$

$$\Rightarrow \begin{cases} -\beta at^2 + t(2\alpha a - \beta b) + \alpha b - \beta c = -t^2 \\ \alpha at^2 + t(2\beta a + \alpha b) + \beta b + \alpha c = 0 \end{cases}$$

$$\begin{cases} -\beta a = -1 \\ 2\alpha a - \beta b = 0 \Rightarrow \beta b = 0 \Rightarrow b = 0 \quad \beta \neq 0 \\ \alpha b - \beta c = 0 \\ \alpha a = 0 \quad \Rightarrow \alpha a = 0 \\ 2\beta a + \alpha b = 0 \quad \Rightarrow a = 0 \end{cases}$$

$y'' = 3$

(E_h) $y'' = 0$ $r^2 = 0$ $r = 0$ racine double

$y_h(x) = (\alpha x + \beta) e^0 = \alpha x + \beta$

Variation de la constante

$y_p(x) = \alpha(x) . x + \beta(x) . 1$

$$\begin{cases} \alpha'(x) x + \beta'(x) . 1 = 0 \\ \alpha'(x) . 1 + \beta'(x) . 0 = 3 \end{cases} \Leftrightarrow \begin{cases} \alpha' + \beta' = 0 \\ \alpha' = 3 \end{cases}$$

$\Rightarrow \alpha(x) = 3x$; $\beta' = -x\alpha' = -3x$

$\alpha(x) = 3x$

$\beta(x) = \frac{-3x^2}{2}$

Donc $y(x) = \alpha x + \beta + x(3x) + \left(\frac{-3x^2}{2}\right)$

$= x^2\left(\frac{-3}{2} + 3\right) + \alpha x + \beta$

$\boxed{y(x) = \frac{3}{2} x^2 + \alpha x + \beta}$

$y'' - 3y' + 2y = t e^t$ (E)

(E_h) : $y'' - 3y' + 2y = 0$

(E.C) : $r^2 - 3r + 2 = 0$

$\Delta = 9 - 8 = 1 > 0$

$r_1 = \frac{3+1}{2} = 2$ $r_2 = \frac{3-1}{2} = 1$

$y_h(x) = \alpha e^{2x} + \beta e^x$

cherchons une sol part.

$y_p(x) = \alpha(x) e^{2x} + \beta(x) e^x$

$$\begin{cases} \alpha'(x) e^{2x} + \beta'(x) e^x = 0 \\ 2\alpha'(x) e^{2x} + \beta'(x) e^x = x e^x \end{cases}$$

$$\Rightarrow \begin{cases} \alpha'(x) e^x + \beta'(x) = 0 \\ 2\alpha'(x) e^x + \beta'(x) = x \end{cases}$$

$\beta'(x) = -e^x \alpha'(x)$

$2\alpha'(x) e^x - e^x \alpha'(x) = x$

$\Rightarrow \alpha'(x) e^x = x \Rightarrow \alpha'(x) = x e^{-x}$

Donc $\alpha(x) = \int x e^{x} dx$

on a : $\alpha(x) = \int x e^{-x} dx = (ax+b) e^{x}$

on cherche a, b tq $((ax+b)e^{-x})' = x e^{-x}$

Donc $x e^{-x} = (a - ax - b) e^{-x}$

$\Rightarrow x = a - b - ax$

$\Rightarrow \begin{cases} a - b = 0 \Rightarrow b = -1 \\ -a = 1 \Rightarrow a = -1 \end{cases}$

Donc $\alpha(x) = (-x - 1) e^{x}$

on a : $\beta'(x) = -\alpha'(x) e^{x}$

$= -((-x-1)e^{-x})' e^{x}$

$= -x e^{-x} e^{x}$

$= -x$

$\Rightarrow \beta(x) = \frac{-x^2}{2}$

Donc la solution de l'eq (E)

$y(x) = y_h(x) + y_p(x)$

$y'' + y = t^2$ (E)

(En) $y'' + y = 0$

(Ec) $r^2 + 1 = 0$; $\Delta < 0$, $r = i$, $r = -i$

$y_h(t) = (\alpha \cos(t) + \beta \sin(t)) e^{0t}$

$\boxed{y_h = \alpha \cos(t) + \beta \sin(t)}$

V.C pour chercher une solution particulière

$y_p(t) = \alpha(t) \cos(t) + \beta(t) \sin(t)$

$\begin{cases} \alpha'(t) \cos(t) + \beta'(t) \sin(t) = 0 \\ -\alpha'(t) \sin(t) + \beta'(t) \cos(t) = t^2 \end{cases}$

$W = \cos^2 t + \sin^2 t = 1$

$\alpha'(t) = \begin{vmatrix} 0 & \sin t \\ t^2 & \cos t \end{vmatrix} = -t^2 \sin t$

$\beta'(t) = \begin{vmatrix} \cos t & 0 \\ -\sin t & t^2 \end{vmatrix} = t^2 \cos t$

Donc

$\alpha(t) = -\int t^2 \sin(t) dt$

$\beta(t) = \int t^2 \cos t \, dt$

$\alpha(t) = \int -t^2 \sin t \, dt$

$= P(t) \sin(t) + Q(t) \cos(t)$

$d^\circ P = d^\circ Q = 2$

$P(t) \sin(t) + Q(t) \cos(t) =$

$(at^2 + bt + c) \sin(t) + (\gamma t^2 + \lambda t + \mu) \cos(t) = F(t)$

on veut $F'(t) = -t^2 \sin t$

$F'(t) = (2at + b) \sin t + (at^2 + bt + c) \cos(t)$

$+ (2\gamma t + \lambda) \cos(t) - (\gamma t^2 + \lambda t + \mu) \sin(t)$

$= -t^2 \sin(t)$

$= \sin(t)(2at + b - \gamma t^2 - \lambda t - \mu) +$

$\cos(t)(at^2 + bt + c + 2\gamma t + \lambda)$

$\begin{cases} 2at + b - \gamma t^2 - \lambda t - \mu = -t^2 \\ at^2 + bt + c + 2\gamma t + \lambda = 0 \end{cases}$

$\begin{cases} -\gamma = -1 \\ 2a - \lambda = 0 \\ b - \mu = 0 \\ a = 0 \\ b + 2\gamma = 0 \\ c + \lambda = 0 \end{cases} \Rightarrow \begin{cases} \gamma = 1 \\ \lambda = 0 \\ \mu = -2 \\ a = 0 \\ b = -2 \\ c = 0 \end{cases}$

Donc : $\boxed{F(t) = -2t \sin(t) + (t^2 - 2) \cos(t) = \alpha(t)}$

$y'' + y' + y = e^{-t} + \cos(t)$ (E)

$\begin{cases} y'' + y' + y = e^{-t} & y_1(x) \\ y'' + y' + y = \cos(t) & y_2(x) \end{cases}$

y_1, y_2 sol (E)

(En) : $y'' + y' + y = 0$

(Ec) $r^2 + r + 1 = 0$ $\Delta = 1 - 4 = -3 < 0$

$r_1 = \frac{-1 + i\sqrt{3}}{2}$; $r_2 = \frac{-1 - i\sqrt{3}}{2}$

Donc $y_h(x) = \left(\alpha \cos\left(\frac{\sqrt{3}}{2} x\right) + \beta \sin\left(\frac{\sqrt{3}}{2} x\right)\right) e^{-\frac{x}{2}}$

$= \alpha y_1(x) + \beta y_2(x)$

① $y'' + y' + y = e^{-t}$ y_p^1

② $y'' + y' + y = \cos(t)$ y_p^2

$y_p^1 + y_p^2$: $y'' + y' + y = e^{-t} + \cos(t)$

$\begin{cases} \alpha'(x) y_1(x) + \beta'(x) y_2(x) = 0 \\ \alpha'(x) y_1'(x) + \beta'(x) y_2' = e^{-x} + \cos(x) \end{cases}$

$ay'' + by' + c = e^{\alpha t}$

$P(t) e^{\alpha t}$; $P(t) = at^2 + bt + c$

$d^\circ P < 2$

6°/ $y'' - y = t\sin(t)$ (E)

Eh $y'' - y = 0$

(Ec) $r^2 - 1 = 0 \Leftrightarrow r^2 = 1$

$r_1 = 1,\ r_2 = -1$

$y_h(t) = \alpha e^t + \beta e^{-t}$

V.C : $y_p(t) = \alpha(t) e^t + \beta(t) e^{-t}$
$= \alpha(t) y_1(t) + \beta(t) y_2(t)$

$$\begin{cases} \alpha'(t) e^t + \beta'(t) e^{-t} = 0 \\ \alpha'(t) e^t - \beta'(t) e^{-t} = t\sin t \end{cases}$$

$W = e^t \cdot e^{-t} - e^t e^{-t} = -2 \neq 0$

$$\alpha'(t) = \frac{\begin{vmatrix} 0 & e^{-t} \\ t\sin(t) & -e^{-t} \end{vmatrix}}{-2} = \frac{-t\sin(t) e^{-t}}{-2} = \frac{t\sin(t) e^{-t}}{2}$$

$$\beta'(t) = \frac{\begin{vmatrix} e^t & 0 \\ e^t & t\sin t \end{vmatrix}}{-2} = -\frac{t\sin(t) e^t}{2}$$

$\alpha(t) = \int \alpha'(t)\,dt$; $\beta(t) = \int \beta'(t)\,dt$

cherchons une primitive de $t\sin(t) e^{\varepsilon t}$ (avec $\varepsilon = \mp 1$)

$\int t\sin(t) e^{\varepsilon t}\,dt = I(t)$

$u' = \sin(t) \longrightarrow u = -\cos(t)$
$v = t e^{\varepsilon t} \longrightarrow v' = (1 + \varepsilon t) e^{\varepsilon t}$

$I(t) = \left(-t\cos(t) e^{\varepsilon t}\right) + \int \left(\cos(t)(1 + \varepsilon t) e^{\varepsilon t}\right)$

$I(t) = -t\cos(t) e^{\varepsilon t} + \int \cos(t) e^{\varepsilon t} + \int \varepsilon t \cos(t) e^{\varepsilon t}$

$\int t\cos(t) e^{\varepsilon t}\,dt = J$

$u' = \cos(t) \longrightarrow u = \sin(t)$
$v = t e^{\varepsilon t} \longrightarrow v' = (1 + \varepsilon t) e^{\varepsilon t}$

$J = t\sin(t) e^{\varepsilon t} - \int (1 + \varepsilon t)\sin(t) e^{\varepsilon t}$

$J = t\sin(t) e^{\varepsilon t} - \int \sin t\, e^{\varepsilon t} - \varepsilon \int t\sin t\, e^{\varepsilon t}$

Donc
$I(t) = \int t\sin(t) e^{\varepsilon t} = -t\cos(t) e^{\varepsilon t} - \int \cos(t) e^{\varepsilon t}$
$+ \varepsilon t\sin(t) e^{\varepsilon t} - \varepsilon \int \sin(t) e^{\varepsilon t}$
$- \varepsilon^2 \int t\sin(t) e^{\varepsilon t}$

$\Rightarrow I(t) = \frac{1}{1+\varepsilon^2}\Big[-t\cos t + \int \cos(t) e^{\varepsilon t} + \varepsilon t\sin t\, e^{\varepsilon t}$
$- \varepsilon\left[\sin(t) e^{\varepsilon t}\right]$

Ex 32

$$\begin{cases} y'' - 3y' + 2y = 0 \\ y(0) = 1,\ y'(0) = 0 \end{cases}$$

Soit (E) : $y'' - 3y' + 2y = 0$

(Ec) $r^2 - 3r + 2 = 0$
$\Delta = 9 - 8 = 1 > 0$
$r_1 = \frac{3+1}{2} = 2$; $r_2 = \frac{3-1}{2} = 1$

Donc $y(x) = \alpha e^{2x} + \beta e^x$

Or on sait que $y(0) = 1$ et $y'(0) = 0$

$\Rightarrow y(0) = \alpha e^0 + \beta e^0 = \alpha + \beta = 1$

$y'(x) = 2\alpha e^{2x} + \beta e^x$

$y'(0) = 2\alpha + \beta = 0$

$$\begin{cases} \alpha + \beta = 1 \Rightarrow \alpha - 2\alpha = 1 \Rightarrow \boxed{\alpha = -1} \\ 2\alpha + \beta = 0 \Rightarrow \beta = -2\alpha \Rightarrow \boxed{\beta = 2} \end{cases}$$

D'où $y(x) = -e^{2x} + 2e^x$

Ex 35 :

$y'' - y = e^x$

$y(0) = 1$ et $y'(0) = 0$

Soit (E) $y'' - y = e^x$
Soit (Eh) $y'' - y = 0$

(Ec) $r^2 - 1 = 0 \Leftrightarrow r^2 = 1 \Rightarrow r = \mp 1$

Donc $y_h(x) = \alpha e^x + \beta e^{-x}$

En cherche une solution particulière :

On a : $y'' - y = e^x$ (polynome de 1er ordre)

On cherche une primitive de la forme :
$(ax^2 + bx + c) e^x = y_p(x)$

$y_p'(x) = (2ax + b + ax^2 + bx + c) e^x$

$y_p''(x) = (2a + 2ax + b + 2ax + b + ax^2$
$+ bx + c) e^x$

$y_p'' - y_p = (2a + 2ax + b + 2ax + b + ax^2$
$+ bx + c - ax^2 - bx - c) e^x$
$= e^x$

$$\begin{cases} 2a + 2a + b - b = 4a = 0 \Rightarrow a = 0 \\ 2a + b + b = 1 \Rightarrow b = 1/2 \end{cases}$$

D'où $y_p(x) = \left(\frac{x}{2} + c\right) e^x$

Donc la solution de (E):

$$y(x) = \alpha e^x + \beta e^{-x} + \left(\frac{x}{2} + c\right) e^x$$

$$y(x) = \alpha e^x + \beta e^{-x} + \frac{x}{2} e^x$$

$c \in \mathbb{R}$ en général on prend $c = 0$

$$= \left(\alpha + \frac{x}{2}\right) e^x + \beta e^{-x}$$

on a: $y(x) = \left(\alpha + \frac{x}{2}\right) e^x + \beta e^{-x}$

$$y'(x) = \left(\frac{1}{2} + \alpha + \frac{x}{2}\right) e^x - \beta e^{-x}$$

$$\begin{cases} y(0) = 1 = \alpha + \beta = \alpha + \alpha + \frac{1}{2} = 2\alpha + \frac{1}{2} \Rightarrow \alpha = \frac{1}{4} \\ y'(0) = 0 = \alpha + \frac{1}{2} - \beta \Rightarrow \beta = \alpha + \frac{1}{2} \Rightarrow \beta = \frac{3}{4} \end{cases}$$

Donc:

$$y(x) = \left(\frac{1}{4} + \frac{x}{2}\right) e^x + \frac{3}{4} e^{-x}$$